Chemical Physics Impact 12 (2026) 101019

Contents lists available at ScienceDirect

Chemical Physics Impact

journal homepage: www.sciencedirect.com/journal/chemical-physics-impact

Full Length Article

Nanoparticle-assisted surfactant formulation for high salinity and temperature conditions: Integrated laboratory evaluation of interfacial properties and wettability modification in carbonates

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ARTICLE INFO

Keywords:
Oil recovery
Surfactant flooding
Interfacial tension
Wettability
Stability
Nanoparticle-assisted surfactant

ABSTRACT

Adsorption and low resistance to high temperature and formation water salinity are among the challenges in applying surfactant-enhanced oil recovery. However, when combined with nanoparticles, surfactants can exhibit improved thermal stability, reduced adsorption onto rock surfaces, and enhanced performance in high-salinity environments. This study goes beyond conventional testing by systematically optimizing a nanoparticle-assisted surfactant solution tailored to the specific high-salinity and high-temperature conditions of the Pilaspi Formation to develop a region-specific EOR solution. It uses crude oil and formation water from the Kurdistan Region of Iraq. A series of experimental tests, including surfactant stability, interfacial tension (IFT), contact angle, spontaneous imbibition and coreflooding were performed at 1800 psia and 80 °C. The IFT, stability and zeta potential results identified that the 0.5 wt% SiO_2/1.5CMC—CTAB formulation was the optimum for IFT reduction and stability, exhibiting a zeta potential of -28 mV. Additionally, it reduced the IFT further by 83% compared to a 67% reduction achieved by the surfactant alone. However, a 1 wt% SiO_2/1.5CMC—CTAB formulation is required to achieve the highest wettability alteration by reducing the contact angles from 170° to 41° Furthermore, spontaneous imbibition tests demonstrated an oil recovery increase from 30.30% using formation water to 74.60% with 1 wt% SiO_2/1.5CMC—CTAB surfactant. Lastly, the coreflooding test showed oil recovery improvements from 39.02% with formation water to 54.27% using 1 wt.% SiO_2/1.5CMC—CTAB surfactant. These findings highlight that the powerful synergistic effect of nanoparticle–surfactant formulation is contingent upon formulation optimization, which remains a critical step to ensure maximum and consistent performance.

1. Introduction

Rising global energy demand underscores the need to maximize hydrocarbon recovery. Enhanced Oil Recovery (EOR) approaches are essential for accessing remaining oil after primary and secondary production [1,2]. Among chemical EOR techniques, surfactant flooding has shown promise for carbonate reservoirs by reducing interfacial tension (IFT) and altering wettability, thereby mobilizing trapped oil. Yet, the effectiveness of conventional surfactants is often limited under harsh reservoir conditions, particularly high temperature and high salinity where IFT reduction and stable displacement fronts are compromised [3,4].

Recent advancements in nanotechnology offer a promising solution to these challenges, as nanoparticles can stabilize emulsions and foams, enhance thermal stability, modify fluid rheology, and improve wettability alteration [5,6]. Nanoparticle (NP)–assisted surfactant flooding has gained increasing attention as an advanced enhanced oil recovery (EOR) technique due to its ability to improve the efficiency of conventional surfactant flooding. The main EOR mechanisms involved include interfacial tension (IFT) reduction, wettability alteration, improved surfactant stability, and mobility control. Nanoparticles can adsorb at the oil–water interface and act synergistically with surfactants to achieve lower and more stable IFT values [3,7–9]. In addition, NPs enhance wettability alteration toward a more water-wet state by forming thin films on rock surfaces, which promotes oil detachment and recovery [10,11]. Nanoparticles also improve surfactant thermal and salinity

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https://doi.org/10.1016/j.chphi.2026.101019
Received 26 November 2025; Received in revised form 31 December 2025; Accepted 2 February 2026
Available online 3 February 2026

tolerance and reduce surfactant loss due to adsorption, making the process more effective under harsh reservoir conditions [12,7,13].

A growing body of work reports synergistic effects from combining surfactants with nanoparticles. For instance, Qiu [14] explored surfactant and nanoparticle-stabilized emulsion flooding for heavy oil recovery in Alaska, identifying an optimized nanoparticle-free micro-emulsion that achieved a remarkably low interfacial tension of 0.08 mN/m. This demonstrated significant potential for enhancing oil recovery, particularly in thermally sensitive reservoirs. Pei et al. [15] further investigated emulsion flooding co-stabilized by silica nanoparticles in watered-out reservoirs, yielding a 50 % incremental oil recovery by diverting flow and improving sweep efficiency. Similarly, Xu et al. [16] assessed the synergistic impact of a surfactant and nanoparticle combination, observing stable emulsions that enhanced sweeping efficiency by diverting flow to low permeability paths. On a molecular scale, Mazyar et al. [17] used molecular dynamics simulation to show that nanoparticles detach oil films, enhance surfactant transport, and alter wettability, contributing to improved recovery. Additionally, Arab et al. [18] found that surface-modified silica nanoparticles can enhance surfactant efficiency for oil emulsification, achieving stable emulsions at lower surfactant concentrations with a 54 % reduction in interfacial tension. Khalafi et al. [19] observed enhanced sweep efficiency and reduced oil blob size in a micro model system, indicating wettability alteration as a dominant mechanism. Notably, Mashat and Abdel-Fattah [2] synthesized nano-surfactant formulations that demonstrated remarkable stability for up to 6 months in high-temperature, high-salinity conditions, achieving reduced interfacial tension and homogeneous emulsions, and suggesting strong potential for oil mobilization in carbonate reservoirs.

Surfactants work by accumulating at the oil-formation water interface, which reduces interfacial tension (IFT) and lowers the capillary forces that trap oil in porous media [20]. However, their performance can be compromised under harsh reservoir conditions. Nanoparticles enhance surfactant efficiency by stabilizing emulsions, improving thermal stability, and contributing to wettability alteration by interacting with rock surfaces and displacing adsorbed oil films [21,22,15].

This study conducts a systematic laboratory investigation to develop and optimize nano-surfactant formulations tailored for the geological conditions of the Kurdistan Region, rather than performing a generic EOR test. The work comprises formulation preparation to core flooding, focusing on their thermal stability and effectiveness in high-salinity carbonate formations. Cetyltrimethylammonium bromide (CTAB) and sodium dodecyl sulfate (SDS) were selected as representative surfactants, while silica and alumina nanoparticles were chosen for their distinct surface chemistries and suitability for high-temperature, high-salinity reservoirs [23–25].

While existing literature has established the general benefits of nano-surfactant EOR, a significant research gap remains in understanding their reservoir-specific optimization for the unique geological conditions of the Pilaspi Formation carbonate reservoirs in the Kurdistan Region. Since optimal formulations vary significantly with reservoir properties, the main aim of this study is to develop a nano-surfactant formulation that is not only effective in general terms but is explicitly optimized for these local conditions [26,27]. The main goal of this experimental study is to provide further insights into the performance of silica and alumina nanoparticles-assisted surfactant flooding. The multi-faceted methodology begins with characterizing the mineralogy and composition of the Pilaspi Formation rock. This is followed by a detailed examination of Interfacial Tension (IFT) and contact angle (CA) tests to elucidate the impact of the formulations on wettability alteration and IFT reduction. Finally, spontaneous imbibition and core flooding tests are employed to quantify the synergistic effect of the nanoparticle/surfactant flooding process on EOR efficiency in real pore spaces, offering a comprehensive assessment toward identifying the optimal formulation for the Kurdistan Region's carbonate reservoirs.

2. Materials

2.1. Rock sample

For this study, carbonate rock from the Pilaspi Formation, a prominent carbonate reservoir in the Kurdistan Region, was used. XRF test (Model: XEPOS) revealed that the rock composition is predominantly calcite with minor traces of quartz (SiO_2) and dolomite ($MgCO_3$) as shown in Fig. 1. This confirms that the rock sample is mainly calcitic carbonate rock.

To facilitate contact angle measurements, thin sections were prepared and dried thoroughly in an oven at 100 °C for 24 h before use. Additionally, core plugs (Table 1) were prepared from the rock samples and dried in an oven at 100 °C for approximately 12 h to remove residual moisture. The core plugs were used for spontaneous and core flooding tests.

2.2. Brine and oil samples

The oil samples and formation water were provided from an oil reservoir in the Kurdistan Region of Iraq. The reservoir brine used in this study was formulated based on formation water ion concentrations presented in Table 2. The mixture was homogenized at 1000 RPM for 24 h to ensure complete homogeneity. On the other hand, the properties of the crude oil used in the study are provided in Table 3.

2.3. Surfactant

Two surfactants, sodium dodecyl sulfate (SDS, anionic) and hexadecyltrimethylammonium bromide (CTAB, cationic), supplied by Merck KGaA, Germany, were utilized and evaluated for their effectiveness in EOR applications in this study. The surfactants' powders were added to the capped jar containing formation brine and the mixture was then stirred for 6 h until complete dissolution was achieved [28].

2.4. Nanoparticle dispersions

The study utilizes aluminum oxide and silicon dioxide nanoparticles. Silica and alumina nanoparticles are commonly used due to their complementary surface charges with typical surfactants and stability in brine [29]. Their availability and proven performance in wettability alteration and EOR make them standard choices [5]. Table 4 shows the properties of the two nanoparticles used.

Different solutions for nanoparticle-surfactants were prepared as shown in Table 5. Silica nanoparticles were chosen to pair with CTAB (cationic surfactant) and alumina nanoparticles with SDS (anionic surfactant) due to their opposite charges, promoting strong interaction [30, 31]. Homogeneous nanoparticle dispersions were prepared using an ultrasound wave maker apparatus. Nanoparticle–surfactant dispersions were prepared using a probe sonicator, where each sample contained the

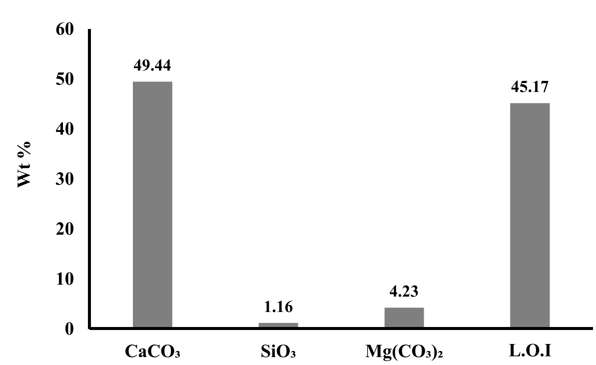

Fig. 1. XRF results of the pilaslpi carbonate rock sample used in the study.

Table 1
Properties of core samples used in this study.

Properties	Core 1	Core 2	Core 3	Core 4	Core 5	Core 6
Length (cm)	7.12 ± 0.15	9.05 ± 0.20	7.47 ± 0.18	5.97 ± 0.12	6.83 ± 0.14	8.24 ± 0.17
Diameter (cm)	3.81 ± 0.01	3.81 ± 0.01	3.81 ± 0.01	3.81 ± 0.01	3.81 ± 0.01	3.81 ± 0.01
Porosity (%)	15.39 ± 0.02	15.12 ± 0.04	15.92 ± 0.04	14.38 ± 0.03	15.46 ± 0.05	15.60 ± 0.03
Liquid permeability (mD)	0.1 ± 0.03	0.13 ± 0.02	0.1 ± 0.02	0.09 ± 0.03	0.11 ± 0.04	0.14 ± 0.02
S_{wi} (%)	34.34 ± 0.32	35.22 ± 0.25	37.20 ± 0.28	38.68 ± 0.17	35.66 ± 0.21	37.72 ± 0.18

selected nanoparticles, surfactant (SDS or CTAB), and 100 mL of formation brine. The probe delivered high-intensity ultrasound (>20 kHz), generating acoustic cavitation that effectively broke nanoparticle aggregates and promoted uniform dispersion in the surfactant–brine matrix [32]). Sonication continued until a homogeneous solution was achieved, indicated by the absence of visible aggregates, after which samples were allowed to equilibrate at room temperature prior to use [31].

3. Methods

The methodology adopted in this study is illustrated in Fig. 2. The experimental workflow outlines the sequential steps followed to optimize and evaluate the nanoparticle–surfactant formulation for enhanced oil recovery. Each step in Fig. 2 is described in detail in the following sections.

3.1. IFT measurement

The IFT between the oil, brine, and different solutions (i.e., surfactant and surfactant-nanoparticle solutions) was measured using an HPHT pendant drop apparatus (model VIT-ES20) as detailed in Table 6. First, the critical micelle concentration (CMC) of CTAB and SDS solutions was determined. CMC is a crucial surfactant property because concentrations above CMC primarily form micelles, and the concentration of surfactant monomers at the interface plateaus, leading to no further IFT reduction [33–35]. Furthermore, interfacial tension tests were conducted for the nanoparticle–surfactant solutions listed in Table 5 to identify the most effective formulations (i.e., yielding the maximum IFT reduction) for both silica and alumina nanoparticles. The tests in Table 6 were performed at ambient pressure.

3.2. Thermal stability and zeta potential tests

The thermal stability of the surfactants and nanoparticle-surfactant solutions was evaluated at 80 °C to evaluate their temperature-dependent stability behavior. Solutions were prepared in Falcon tubes, sealed tightly, and placed in an oven for 5, 10, 15, and 20 days. IFT measurements were conducted after each aging period to assess the thermal stability. Limestone samples were ground and sieved to obtain particles below 40 μm for better dispersion homogeneity. The powder was then washed with deionized water using an ultrasonic wave maker for 20 min, followed by drying in an oven at 80 °C for 24 h. Subsequently, 0.2 *g* of limestone powder was mixed with 20 mL of the solution (surfactant or nanoparticle-surfactant) at 80 °C for 3 days under sonication. Moreover, zeta potential measurements were performed using a Malvern Zetasizer (Nano ZS Zen 3600) at room temperature to examine the stability of particles in dispersion. Zeta potential reflects the magnitude of repulsion between particles in a suspension. Higher values indicate greater electrical stability, while lower values suggest a tendency to aggregate [36,37].

Zeta potential tests on rock powder alone provide information about the natural surface charge and wettability of the rock in a given brine. Likewise, zeta potential measurements of nanofluids alone are mainly used to evaluate particle stability, where higher absolute zeta potential values indicate better dispersion due to stronger electrostatic repulsion [38,39]. However, once nanoparticles are injected into a reservoir, their performance is not determined only by their stability in the fluid, but also by how they interact with the rock surface. Conducting zeta potential measurements with both rock powder and nanoparticles together makes it possible to directly observe nanoparticle adsorption, changes in rock surface charge, and electrostatic interactions at the rock–nanoparticle interface, which are critical for understanding wettability alteration [10].

3.3. Contact angle measurement

The contact angle between various fluids (listed in Table 5) and oil-aged thin section surfaces were measured using a sessile drop method with an optical goniometer contact angle [40]. A drop of each solution

Table 3
Properties of the crude oil used in the study.

API	Density (g/cc) at ambient conditions	Density (g/cc) at ambient pressure and 80 °C	Viscosity (cp)
36.6	0.745	0.8	9.5

Table 4
Properties of aluminum oxide and silicon dioxide nanoparticles.

Nanoparticle	Weight (g)	Purity (%)	Average Particle Size (nm)	Specific surface area (m^2/g)
Silicon Dioxide	25	99.9	30	> 150
Gamma Alumina	25	99	20	< 200

Table 5
Nanoparticle-surfactant solutions used in the study.

Suspension Name	Concentration (wt %)			
	CTAB surfactant *	SDS surfactant**	Silica Nanoparticle	Alumina Nanoparticle
SNP1	0.135	—-	0.10	—-
SNP2	0.135	—-	0.25	—-
SNP3	0.135	—-	0.5	—-
SNP4	0.135	—-	1.0	—-
SNP5	—-	0.15	—-	0.10
SNP6	—-	0.15	—-	0.25
SNP7	—-	0.15	—-	0.5
SNP8	—-	0.15	—-	1.0

(*) This is 1.5CMC concentration for CTAB surfactant.
(**) This is 1.5CMC concentration for SDS surfactant.

Table 2
Composition of reservoir formation water used to prepare the brine used in the study.

Ion	Cl^-	SO_4^{2-}	S^{2-}	HCO_3^-	Ba^{2+}	Na^+	K^+	Cu^{2+}	Mg^{2+}	Ca^{2+}	Total
mg/L	16,200	1141	2750	1600	67	10,789	1610	171	380	1520	34,461.6

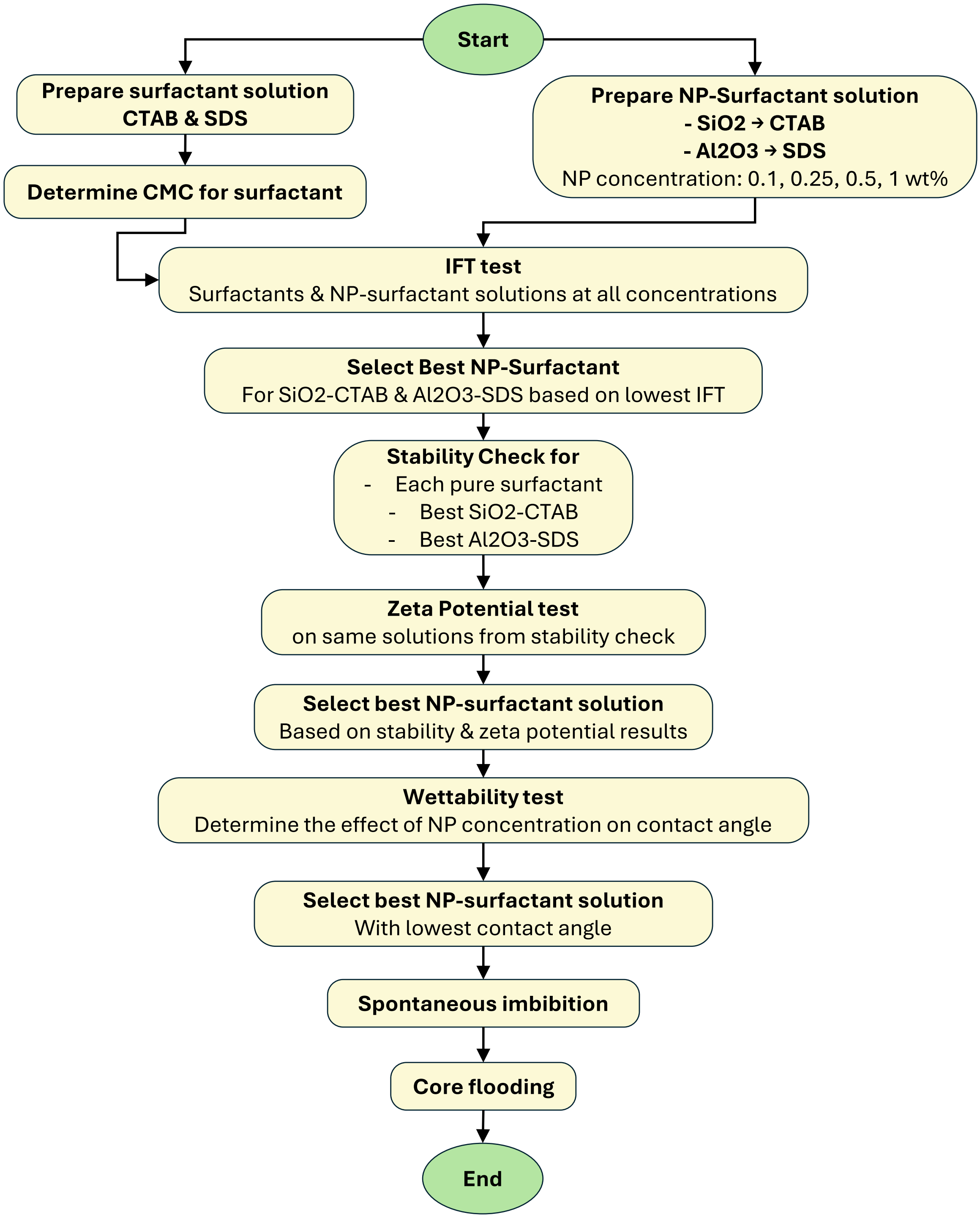

Fig. 2. Experimental workflow for optimization and evaluation of nanoparticle-surfactant formulations.

was placed on a thin section representing the rock surface. The contact angle was captured by the camera and measured by the software over 48, 120, 240, and 480 h after aging at 1800 psia and 80 °C. Nevertheless, the measured contact angles were used to determine the Wettability Alteration Index (WAI) which reflects the degree of wettability alteration using Eq. (1) [41]:

$$WAI = \frac{\theta_{original} - \theta_{final}}{\theta_{original} - \theta_{initial}} \tag{1}$$

Where:

$\theta_{original}$ is the original contact angle measured after aging the thin sections with crude oil, θ_{final} is the final contact angle after the wettability alteration. WAI results determined the optimal nanoparticle concentrations for both imbibition and core flooding tests.

3.4. Spontaneous imbibition and core flooding test

Spontaneous imbibition tests were performed on core plugs 2 through 6 to evaluate the effectiveness of various solutions in displacing residual oil (Table 7). These solutions included formation water (brine), standalone surfactants (SDS and CTAB), and combinations of nanoparticles with surfactants (silica-CTAB and alumina-SDS) chosen based

Table 6
Oil-brine interfacial tension tests for different solutions at 100 psia and 80 °C.

Test number	Solution	Objective
1	CTAB surfactant	Determine the critical micelles concentration
2	SDS surfactant	
3	CTAB surfactant (1.5CMC)	Determine the effective silica nanoparticle concentration in CTAB(1.5CMC)
4	SNP1	
5	SNP2	
6	SNP3	
7	SNP4	
8	SDS surfactant (1.5CMC)	Determine the effective alumina nanoparticle concentration in SDS(1.5CMC)
9	SNP5	
10	SNP6	
11	SNP7	
12	SNP8	

on the results of IFT and contact angle measurements. The tests aimed to verify the wettability alteration and IFT reduction achieved by these treatments. An Amott cell was used to conduct five spontaneous imbibition tests at 80 °C and ambient pressure. Each test measured the oil recovery rate over a period of approximately 25 days. Afterward, a core flooding test was conducted using a core flooding apparatus and core plug 1 to evaluate oil recovery efficiency under reservoir conditions of 80 °C and 1800 psia. The injection rate during the test was maintained at 0.1 mL/min.

4. Results

4.1. Effect of surfactant on IFT

First, the synthesized formation brine-oil interfacial tension was determined and was 12.45 mN/m. Afterward, the interfacial tension between the oil and solutions of CTAB and SDS surfactants at different concentrations were measured at 1800 psia and 80 °C to determine the CMC concentration for each surfactant. As shown in Fig. 3, the CMC of the SDS-oil system was 0.1 wt %, which corresponds to an IFT of 5.95 mN/m. On the other hand, 0.09 wt % of CTAB was observed as the CMC of the CTAB-oil system which yields an IFT of 4.1 mN/m. This result indicates that CTAB is more effective in lowering the IFT. However, to compensate for the loss of surfactants within the porous media, the CMC for each system was multiplied by 1.5 [42]. The concentration of 1.5CMC is considered the optimum and is used for contact angle measurements.

4.2. Effect of coupled nanoparticle-surfactant on IFT

Different concentrations (0.1–1 wt %) of silica and alumina nanoparticles, as explained in Table 5, were added to (1.5CMC) of both CTAB and SDS surfactant solutions respectively. The IFT results in Fig. 4 show that silica-CTAB solution results in more IFT reduction than alumina-SDS solution. The solution of 1.5CMC—CTAB containing 0.5 wt % of

Table 7
Summary of the spontaneous imbibition and core flooding tests.

Test	Solution	Core plug	Test Type	Pressure and Temperature
1	Formation water	2	Spontaneous imbibition	80 °C and 1800 psia
2	CTAB (1.5CMC)	3		
3	SDS (1.5SDS)	4		
4	SNP3	5		
5	SNP6	6		
6	SNP4	1	Core flooding	

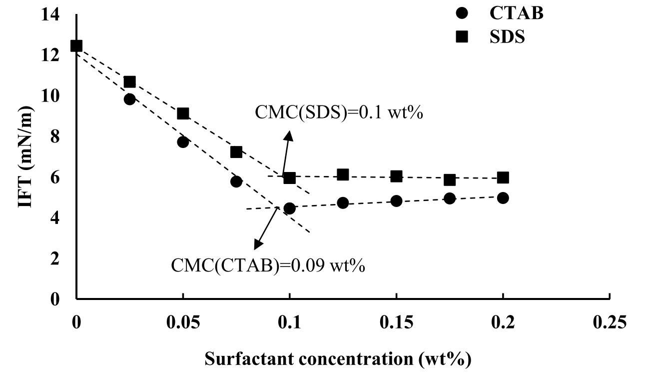

Fig. 3. IFT between crude oil, CTAB and SDS surfactants at different concentrations at 1800 psia and 80 °C.

silica nanoparticle (SNP3) lowered the IFT to 2.11 mN/m. In contrast, the lowest IFT of 4.03 mN/m was achieved when 0.25 wt % of alumina was added to 1.5CMC-SDS solution (SNP6). However, increasing the nanoparticle concentration beyond 0.5 wt % for silica and 0.25 wt % for alumina could not further decrease the IFT. This indicates that the concentrations of 0.5 wt % and 0.25 wt % are the effective concentrations of silica and alumina nanoparticles respectively.

In general, both nanoparticles contributed positively to further lowering the interfacial tension by enhancing the stability of the surfactants. However, the silica-CTAB combination demonstrated the best performance in IFT reduction is attributed to the adsorption of the nanoparticle-surfactant synergy on the water-oil interface. In this mechanism, CTAB molecules electrostatically adsorb onto negatively charged silica, resulting in amphiphilic complexes that preferentially accumulate at the interface and further reduce interfacial free energy compared to surfactant alone [43]. These findings align well with the literature [29,44].

4.3. Thermal stability and zeta potential tests

Stability tests were conducted on surfactant solutions (CTAB and SDS) at a concentration of 1.5 times their CMC, and on nanoparticle-surfactant solutions (silica-CTAB and alumina-SDS) at specific nanoparticle concentrations (0.5 wt % silica and 0.25 wt % alumina) at 80 °C. The selected concentrations for both nanoparticles were selected based on their performance in IFT reduction. In general, IFT values for both

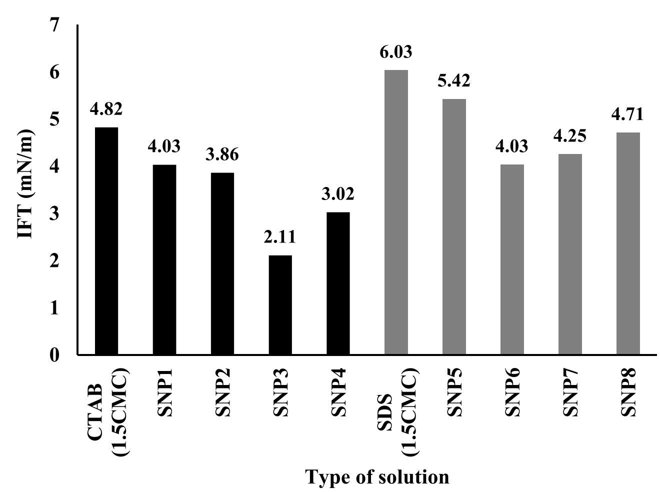

Fig. 4. Interfacial tension between oil and CTAB (1.5CMC), SDS(1.5CMC), silica-CTAB(1.5CMC) and alumina-SDS(1.5CMC) at different concentrations of silica and alumina nanoparticles (Table 5) at 1800 psia and 80 °C. The results are expressed as means ± SD ($n = 3$).

CTAB and SDS surfactants increased slightly with time as seen in Fig. 5. However, CTAB (1.5CMC) solution exhibited more stability than the SDS(1.5CMC) solution after 240 h. The increase in the IFT values can be attributed to the presence of the divalent ions that would exacerbate their stability. Divalent ions screen the electrostatic double layer and can precipitate or be complexed with anionic surfactants like SDS, reducing their interfacial activity [45]. Similarly, even cationic surfactants such as CTAB are sensitive to multivalent counterions, which alter adsorption behavior and may slightly diminish long-term stability [46]. CTAB exhibited greater stability than SDS in brine rich in divalent ions because its positively charged quaternary ammonium headgroups are not prone to complexation or precipitation with Ca^{2+} and Mg^{2+}, enabling sustained micelle and interfacial activity over time. In contrast, SDS (anionic) readily forms insoluble complexes or precipitates with divalent cations, leading to destabilize the micelle and reduce interfacial effectiveness over extended durations [47].

Interestingly, adding nanoparticles significantly improved the stability of both surfactant solutions as shown in Fig. 5. The IFT of the silica-CTAB (SFNP3) solution only changed from 2.11 to 2.25 mN/m after aging, which can be related to the thermal resilience of silica nanoparticles under high-salinity and elevated temperature conditions [21,48]. Likewise, alumina nanoparticles enhanced the thermal stability of the SDS solution. This improved stability could be one factor contributing to the better oil recovery observed with nanoparticle-surfactant synergies compared to standalone surfactant solutions via IFT reduction. This will be discussed in more detail in the next sections [30,49].

In the same direction, zeta potential results in Table 8 indicates that SNP3 solution exhibited the highest zeta potential (−28 mV), indicating greater electrical stability, which suggests a lower tendency to aggregate [36,37]. This supports our stability test findings where SNP3 was the most stable formulation observed. Lower zeta potential values in other solutions likely stem from the presence of divalent ions (e.g., calcium) that weaken repulsion and promote aggregation [30,49].

4.4. Wettability alteration by surfactant, and coupled nanoparticle-surfactant solutions

The contact angles of crude oil on the oil-aged thin sections were measured at 1800 psia and 80 °C to be nearly 169°, indicating an oil-wet state. To alter the wettability towards water-wet, the thin sections were treated with surfactants (CTAB and SDS), and nanoparticles (alumina and silica)-surfactant combination. CTAB and SDS surfactants at their 1.5CMC reduce the contact angle to approximately 83° and 134°, respectively as shown in Fig. 6.

Aluminum oxide (Al_2O_3) and silicon dioxide (SiO_2) nanoparticles have been extensively studied for their ability to change the wettability of reservoir rocks and improve oil recovery. When injected into porous rocks as nanofluids, these nanoparticles attach to the mineral surfaces, reducing the affinity between oil and rock and shifting the wettability toward a more water-wet state. This change enhances capillary-driven water imbibition, which in turn improves oil displacement during flooding processes [50]. The underlying mechanism behind these changes is the adsorption of nanoparticles at the solid–fluid interfaces. This adsorption modifies the surface energy and the structure of rock–fluid contacts, decreasing oil adhesion and promoting water wetness [51,50]. The extent of wettability alteration depends on factors such as nanoparticle concentration, temperature, and reservoir brine composition, with optimized conditions leading to more significant shifts in contact angle and potential increases in oil recovery [51].

Table 8
Zeta potential measurements for surfactant solutions and surfactant-nanoparticle combinations at room temperature.

Solution	Concentration	Makeup water	Zeta potential (mV)
FW	—	Formation brine of the oil reservoir	−3.5 ± 0.53
CTAB	1.5 CMC		−2.5 ± 0.38
SDS	1.5 CMC		−6 ± 0.56
CTAB+Silica NP	1.5 CMC + 0.5 wt %		−28 ± 0.68
SDS+Alumina NP	1.5 CMC + 0.25 wt %		−5 ± 0.44

CTAB is more effective than SDS in reversing the thin sections' oil-wet state which supports the findings of the IFT tests. This is because CTAB molecules have a positive charge, like the charge of the carbonate rock surface. When thin sections are treated with CTAB, the CTAB molecules absorb on the rock surface and displace the negatively charged carboxylic components through ion-pair interactions. This would ultimately eject the trapped oil through water imbibition into the pores [52–54].

Adding alumina nanoparticles at various concentrations to SDS could not significantly improve their performance. For instance, the contact angle on oil-aged thin sections treated with 0.25 wt. % alumina nanoparticles and 1.5 CMC SDS decreased from 169° to around 150° This minimal change suggests that alumina nanoparticles offered little improvement in altering wettability compared to using SDS alone (134°). In contrast, the addition of 0.25 wt % silica nanoparticles to the 1.5 CMC CTAB solution resulted in a smaller reduction in contact angle (from 163° to 119°) compared to CTAB alone (from 169° to 83°). However, increasing the silica concentration to 0.5 and 1 wt % significantly enhanced the performance of CTAB, reducing the contact angle to 55° and 41°, respectively. This improvement is attributed to the synergy between adsorption and disjoining pressure, which is believed to be the primary reason for the significant improvement in wettability observed with the silica NP-CTAB combination [6].

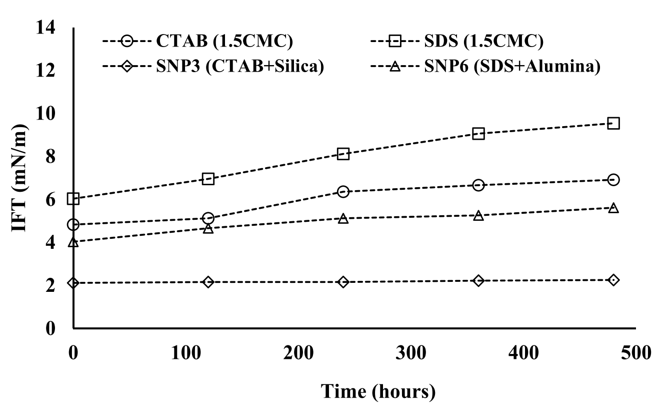

Fig. 5. Interfacial tension of oil with different standalone surfactants and surfactant-nanoparticle solutions as a function of time, in 1800 psia and 80 °C.

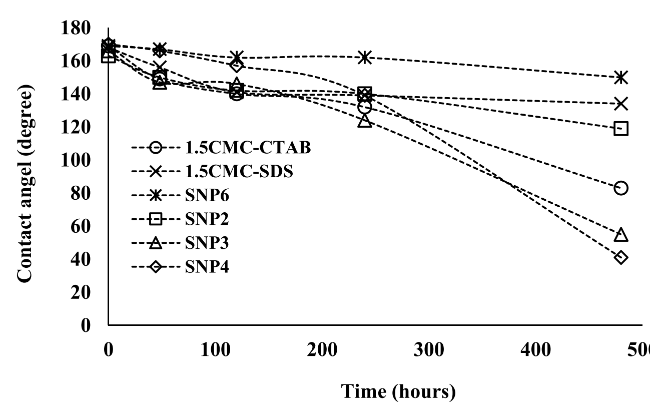

Fig. 6. Average contact angle results for surfactant solutions and surfactant-nanoparticle combinations over 480 h at 1800 psia and 80 °C.

Although 0.5 wt % SiO_2–CTAB provided the most stable dispersion and optimal IFT reduction-evidenced by a robust negative zeta potential, higher loading at 1 wt % delivered superior wettability alteration, as greater particle mass enabled more extensive surface coverage, masking hydrophobic residues and forming a hydrophilic coating. This illustrates the well-documented trade-off between colloidal stability and the efficacy of surface modification [55].

Furthermore, the Wettability Alteration Index (WAI) was determined using the contact angles of 0.25 wt % alumina–SDS and silica–CTAB at 0.25, 0.5, and 1 wt % silica. As shown in Fig. 7, the highest wettability alteration was achieved with the 1 wt % silica–CTAB formulation, supporting the contact angle results.

4.5. Spontaneous imbibition oil recovery

Spontaneous imbibition tests were performed on core plugs 2 through 6 to evaluate the effectiveness of various solutions in displacing oil in the oil-aged core plugs at ambient pressure and 80 °C. These solutions included formation water (FW), standalone surfactants (SDS and CTAB), and combinations of nanoparticles with surfactants (silica NP/CTAB and alumina NP/SDS). The tests aimed to verify the wettability alteration and IFT reduction achieved by these treatments. As shown in Fig. 8, formation water imbibition was the first test conducted on core plug 2, serving as a baseline for comparison. This test resulted in an oil recovery of 30.30 %. Afterward, 1.5CMC—CTAB and 1.5CMC-SDS solutions were imbibed in core plugs 3 and 4 respectively. Both solutions improved the oil displacement and recovery compared to formation water. 1.5CMC—CTAB yielded 5.18 mL of oil (60.86 %) whereas the 1.5CMC-SDS displaced 2.16 mL of oil (35.98 %). These findings corroborate the contact angle and IFT results, where the CTAB-based system displayed superior performance, which explains the higher oil recovery observed for CTAB compared to SDS.

To improve the performance of surfactants, 0.25 wt % alumina and 1 wt % silica nanoparticles were added to 1.5CMC-SDS and 1.5CMC—CTAB solutions respectively. The results showed that the silica-CTAB formulation recovered 74.6 % oil and alumina-SDS formulation yielded 39.68 %. Silica-CTAB performed better, likely due to better surfactant transport through the pores and reduced adsorption on the rock surface. The combined effects of the silica/CTAB solution offer better surfactant transport through the pores and reduced adsorption on the rock surface. Furthermore, the enhanced recovery is primarily attributed to surfactant-driven wettability alteration and interfacial tension reduction, which reduce capillary pressure and thus improve oil displacement efficiency in the pores [56,57].

The results from Figs. 4 and 6 revealed that a 0.5 wt % SiO_2-CTAB mixture was more effective in reducing interfacial tension (IFT), while a 1 wt % mixture was required to achieve a lower contact angle, indicating improved wettability. This difference highlights that IFT reduction is governed by interfacial adsorption, which can be achieved at lower nanoparticle concentrations, whereas wettability alteration depends on sufficient surface coverage and modification of the rock, requiring higher concentrations [58]. Additionally, the results in Fig. 6 confirmed that 1 wt % SiO_2 led to higher oil recovery, emphasizing the critical role of wettability in enhancing oil mobilization.

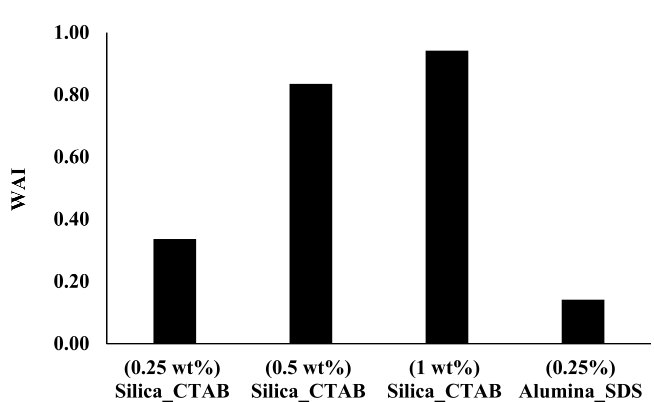

Fig. 7. Wettability alteration index measured based on the contact angles for NP-surfactant solutions of SNP2, SNP3, SNP4, and SNP6.

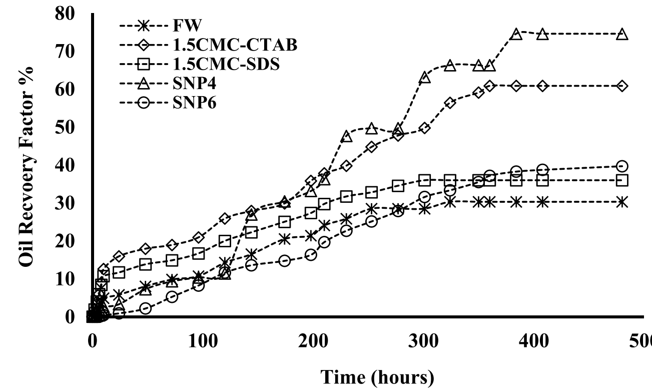

Fig. 8. Recovery factor measured by the spontaneous imbibition method using formation water, surfactant solutions, and surfactant-nanoparticle combinations over time. The data are presented for core plugs 2 through 6 used in the tests at ambient pressure and 80 °C.

To confirm the role of wettability and interfacial tension (IFT) on the oil recovery factor during the spontaneous imbibition test, contact angle and IFT data were plotted against the corresponding recovery factors, as shown in Fig. 9. The results demonstrate a strong linear correlation between contact angle and recovery factor, suggesting a steady influence of wettability on oil recovery. On the other hand, the recovery factor exhibits an exponential relationship with IFT, indicating that IFT plays a critical role at low values and has a compounding effect on oil recovery as it drops.

4.6. Core flooding tests

The core flooding experiment was performed on core plug 1, which was initially saturated with crude oil. The test was performed for 48 h. During the test, pressure and temperature were maintained at 1800 psi and 80 °C respectively, and the injection rate for all fluids was 0.1 mL/min. The core plug was then sequentially flooded with formation brine, 1.5CMC—CTAB solution, and 1 wt %-silica/1.5CMC—CTAB (SNP4) solution as shown in Fig. 10. This specific combination was selected based on its superior performance in wettability alteration and spontaneous imbibition tests as discussed in Sections 4.4 and 4.5. The formation water yielded 3.2 mL of oil (39.02 %). Subsequent injection of 1.5 CMC—CTAB solution increased the recovered oil recovery to 47.56 % whereas SNP4 resulted in 54.27 % oil recovery as shown in Fig. 8. Accordingly, the recovery increments yielded were 8.54 % and 6.71 % by standalone surfactant and SNP4 solutions respectively. The core flooding results confirm the critical role of wettability alteration in governing oil displacement, thereby enhancing oil recovery in complex porous media. As observed in Figs. 8 and 10, the higher recovery factor was obtained during the spontaneous imbibition test for the same SNP4 solution. The higher oil recovery observed in spontaneous imbibition compared to core flooding using the same solutions can be attributed to the dominance of capillary forces in the former, which become more effective under water-wet conditions induced by the surfactant and nanoparticle treatments [59,60].

Like the spontaneous imbibition test, contact angle and IFT demonstrated a comparable relationship with the recovery factor during the core flooding test (Fig. 11). The contact angles exhibited a steady linear correlation with oil recovery, indicating that more water-wet conditions enhance recovery. In contrast, IFT showed an exponential correlation

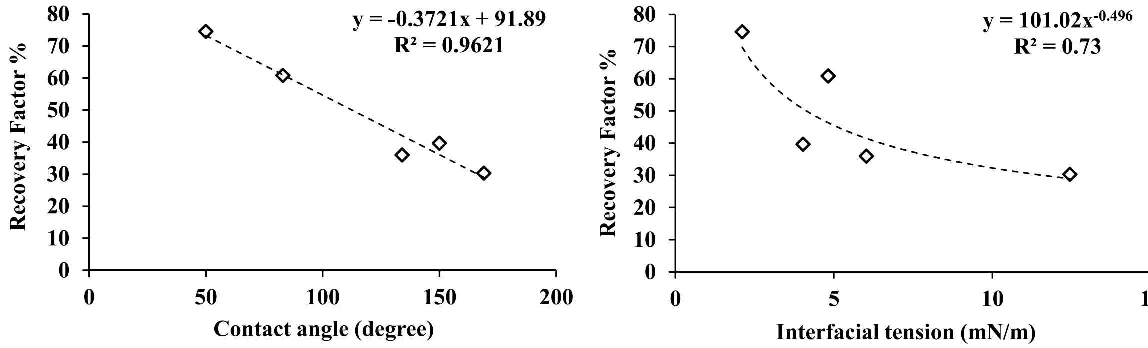

Fig. 9. Recovery factors measured by the spontaneous imbibition method as functions of measured contact angles (left) and interfacial tensions (right) using formation water, 1.5CMC—CTAB, 1.5CMC-SDS, SNP4, and SNP6 solutions.

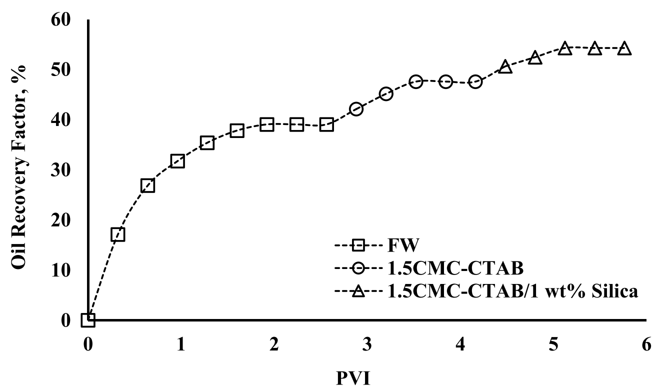

Fig. 10. Oil recovery factor measured by the coreflooding method during formation water injection, 1.5CMC—CTAB solution injection, and 1.5 CMC—CTAB surfactant and 1 wt % of Silica nanoparticles solution at 80 °C and 1800 psi.

with recovery, highlighting its pronounced effect at lower values.

5. Conclusion

This study evaluated the potential of optimized nanoparticle (silica and alumina)-assisted surfactant (CTAB and SDS) flooding using carbonate rock from the Pilaspi Formation, formation water, and crude oil from the Kurdistan Region under reservoir conditions (80 °C and 1800 psia). CTAB outperformed SDS in lowering interfacial tension (12.45 to 4.82 mN/m). The optimal nanoparticles-surfactant concentrations for IFT reduction were 0.5 wt % SiO_2-CTAB and 0.25 wt % Al_2O_3-SDS. Wettability alteration required higher silica nanoparticles loading, 1 wt % SiO_2–CTAB, to achieve the lowest contact angle (170° to 41°), shifting the carbonate surface toward strong water-wetness. This higher concentration likely provided greater surface coverage of nanoparticles on the rock, amplifying the wettability alteration effect, even though the 0.5 wt % solution was more stable in suspension. This suggests that 0.5 wt % is optimal for stability and IFT reduction, whereas 1 wt % is optimal for wettability alteration and oil recovery. Spontaneous imbibition and core flooding confirmed the critical role of wettability alteration in recovery performance, with 1 wt % SiO_2-CTAB yielding the highest oil recovery of 74.6 % and 54.27 % in both tests respectively. Recovery factor showed a strong linear correlation with wettability and an exponential correlation with IFT, highlighting the dominant role of wettability in displacement efficiency and the critical importance of IFT reduction at low values.

Overall, the results demonstrate that Silica nanoparticles-assisted CTAB flooding can significantly enhance oil recovery in carbonate reservoirs under high salinity and temperature through the synergistic effects of IFT reduction and wettability alteration, with findings applicable to field-scale EOR operations. However, achieving the powerful synergistic effect of nanoparticle–surfactant systems require optimized formulations to deliver effective performance.

CRediT authorship contribution statement

Ararat Rahimy: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Conceptualization. **Ramyar Suramairy:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology, Data curation. **Dana A. Hasan:** Writing – review & editing, Visualization, Resources. **Amanj W. Khalid:** Writing – review & editing, Software, Resources.

Declaration of competing interest

The authors declare that they have no competing financial interests or personal relationships between co-authors that could have appeared

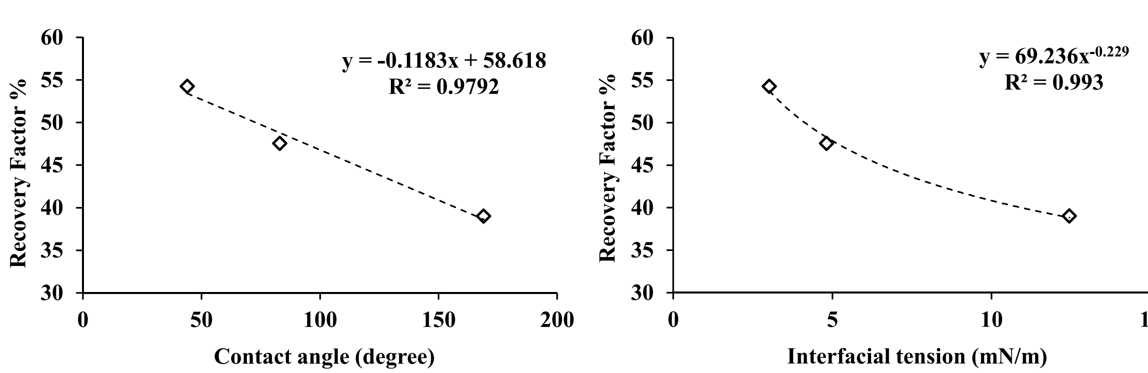

Fig. 11. Recovery factors measured by the core flooding method as functions of measured contact angles (left) and interfacial tensions (right) using formation water, 1.5CMC—CTAB, 1.5CMC-SDS, SNP3, and SNP6 solutions.

to influence work reported in this review paper.

Data availability

Data will be made available on request.

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